

HOJUNE KIM

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EDUCATION

Stanford University

M.S. in Aeronautics and Astronautics

- GPA: 4.0/4.0(Overall)

CA, United States

09/2025 – 03/2027 (Expected)

Seoul National University (SNU)

B.S. in Aerospace Engineering, summa cum laude

- GPA: 4.0/4.0(Major), 3.91/4.00(Overall)

Seoul, South Korea

03/2019 – 08/2025

* 1.5-year absence from military service

WORK EXPERIENCE

Foundation Robotics - VLA team

Robotics Software Intern

- Developing a generalized teleop-free loco-manipulation VLA training pipeline using world models, extending the embodied Gaussian Splatting approach from the LEGS paper to broader scenes and embodiments

CA, United States

06/2026 – Present

German Aerospace Center (DLR) - Humanoid TORO team

Guest Researcher, Advisor: Prof. Jinoh Lee

- Implemented an adaptive locomotion system using semantic SLAM to optimize 8 gait parameters on uneven and deformable terrains via motion planning based on Divergent Components of Motion
- Developed a scalable semantic segmentation model in Jetson Orin AGX by fusing a stream-based active learning classifier with the Segmentation Anything Model, training it via a human-robot interaction loop using a HoloLens
- Improved resilience to common failures (e.g., contact loss, localization glitches) by implementing a Kalman Filter for robust visual-proprioceptive odometry, winning 1st place in the IEEE-RAS Humanoids competition

Munich, Germany

09/2024 – 02/2025

RESEARCH EXPERIENCE

Stanford University - Multi-robot Systems Lab

Research Intern, Advisor: Prof. Mac Schwager

- Designed a Gaussian Splatting simulator for loco-manipulation, enabling automated large-scale data generation
- Developing a zero-shot visuomotor policy for humanoids, eliminating dependence on human demonstrations

CA, United States

09/2025 – Present

ETH Zurich

Robotic Systems Lab, Research Intern, Advisor: Prof. Cesar Cadena

- Developed and trained reinforcement learning policies using a knowledge distillation framework for crowded navigation on the quadruped robot ANYmal, achieving a 74.5% success rate on static and dynamic terrains
- Utilized PPO in Isaac Lab for training and implemented a Gated Recurrent Unit to handle dynamic obstacles

Zurich, Switzerland

02/2024 – 06/2024

Vision for Robotics Lab, Research Intern, Advisor: Prof. Margarita Chli

- Developed a symbolic Gaussian Belief Propagation (GBP) solver in C++ for continuous-time SLAM, achieving a 2.4x to 110.3x speedup compared to conventional methods
- Benchmarked the performance of distributed optimization GBP against the centralized NLLS solver (Ceres) for both continuous- and discrete-time SLAM using hand-achieved data

02/2024 – 08/2024

Seoul National University - Robust Perception and Mobile Robotics Lab

Research Intern, Advisor: Prof. Ayoung Kim

- Proposed a novel 2D odometry method with only mmWave radar, achieving an 85-90% reduction in RPE on indoor datasets by using a bidirectional weighted ICP algorithm
- Designed a handheld multi-sensor system and held camera-LiDAR-radar joint calibration via graph optimization

Seoul, South Korea

02/2023 – 01/2024

PUBLICATIONS

- H. Kim**, T. Chen, J. Sun, L. W. Osterberg, Q. Chen, K. Wang, M. Schwager. LEGS: Fine-Tuning Teleop-Free VLAs for Humanoid Loco-manipulation in an Embodied Gaussian Splatting World. *Under review at CoRL 2026*. arXiv:2606.01458.
- H. Kim**, H. Jang, A. Kim. 2D Ego-Motion with Yaw Estimation using Only mmWave Radars via Two-Way weighted ICP. *IEEE ICRA 2024 Workshop on Radar in Robotics*, Yokohama, 2024.
- H. Kim***, J. Lee*, G. Mesesan, R. Giubilato, R. Schuller, K. Fründ, J. Lee. Knowledge-enabled Adaptive Locomotion Using Semantic SLAM with Active Learning. *In Progress*.

SKILLS

Platforms: Humanoids, ANYmal, GO2, Drone, Vehicle

Frameworks: ROS1/2, Isaac Sim/Lab, Pytorch, PX4, Gazebo

Programming: C/C++, Python, MATLAB

Tools: Cmake, AWS EC2, Docker, Git, CAD

AWARDS

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| 1st Place , IEEE-RAS Humanoids Competition Adult-sized Biped Free Walk | 11/2024 |
| Minister of National Defense Award (out of 953 teams), Korea Defense Startup Challenge | 12/2022 |
| Gold Prize (out of 60 teams), International Student Car Competition Autonomous Driving Sector | 10/2021 |
| SNU Alumni Association Chairman's Award (Sole recipient among the College of Engineering) | 08/2025 |
| • Awarded for exceptional leadership and contributions to the university community | |

PATENTS

1. Parking Location Tracking System, *KR102291377B1*, 2021
Hojune Kim, Taekin Kim, Jinhwan Na, Jaeyoung Lee, Seunghwan Jeong
2. Document editing system supporting online document creation with search function, *KR100143446*, 2022
Hojune Kim, Jinsu Choi

EXTRA CURRICULAR

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|---|-------------------|
| Dentronic
<i>Co-Founder</i> | 07/2024 – 12/2025 |
| • Built a back-drivable robotic assistant arm designed to automate suction and retraction for solo dentist procedures | |
| • Generated \$125k in revenue from the first product batch in a single day, serving customers across the US and Korea | |
| Fire Butterfly , Autonomous Drone Association
<i>Lead Software Engineer</i> | 02/2023 – 08/2023 |
| • Engineered auto-landing system on drone with sub-5cm landing accuracy using C2 Bézier curve motion planning | |
| • Validated the system in both Gazebo simulation and PX4 on-board flight with Jetson Xavier, leading to a technology transfer to a private company | |
| Army Aviation Operations Command , Republic of Korea Army
<i>CH-47D Helicopter Flight Attendant, Sergeant</i> | 08/2021 – 02/2023 |
| • Conducted maintenance on CH-47D helicopters and participated in tactical missions, including over 60 flight hours dedicated to two major forest fire extinguishment operations (2022 Uljin, 2022 Yeongdeok) | |
| AI Tech Play , Non-Profit Organization for AI education
<i>Co-organizer & Hardware Leader</i> | 02/2021 – 08/2021 |
| • Collaborated with MIT Lincoln Lab to host AI camp and RC-car competition for over 200 nationwide children, providing RC-car simulation tools and handmade platform to students | |
| • Led a team to develop a multi-modal, ROS2-controlled RC-car with Jetson Nano, creating 7 Python tutorials on autonomous driving | |
| SNU ZERO , Autonomous Driving Car Association
<i>Sensor Technician</i> | 01/2021 – 08/2021 |
| • Designed the autonomous vehicle system from chassis, developing GNSS-aided inertial navigation system via Extended-Kalman Filter assisted by line detection for robust localization | |
| • Engineered a multi-modal perception system for dynamic and static obstacle avoidance, using camera-LiDAR fusion | |

SCHOLARSHIPS

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| Chungsoo Overseas Scholarship Chungsoo Scholarship Foundation | 2025 – 2027 |
| Korea-Germany Junior Research Fellowship Support Max Planck POSTECH/KOREA | 09/2024 |
| Global Leadership Program Scholarship SNU | 02/2024 |
| Kwanjeong Undergraduate Scholarship Kwanjeong Educational Foundation | 2021 – 2024 |